

Shaheer Muzaffar Khan

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🌐 Shaheer Muzaffar Khan

📍 Islamabad, Pakistan

Professional Summary

Dedicated Artificial Intelligence undergraduate (4th Semester) at NUST with a strong foundation in Machine Learning, Data Structures and Algorithms, and AI pipeline design. Experienced in taking projects from concept to completion, including the successful delivery of an AI-powered, full-stack application. Passionate about building robust, user-focused systems and integrating machine learning models into practical solutions. A collaborative team player with a proactive approach to complex technical challenges.

Projects

Jholly Phonics: AI-Powered Pronunciation App

- **Developed** a mobile application using React Native (Expo), designing an intuitive UI and implementing seamless audio upload workflows for real-time phoneme classification.
- **Engineered** the backend architecture using FastAPI to successfully integrate the frontend with a Whisper-based feature extractor and BiLSTM classifier developed by the AI team.
- **Collaborated** in a cross-functional academic environment to deliver an interactive educational tool that provides users with instant phoneme predictions and real-time accuracy feedback.

SmartPulse: AI-Driven Business Management System

- **Designed** the comprehensive UI/UX and implemented the complete frontend architecture using React, delivering intuitive modules for customer profiles, analytics, and settings.
- **Engineered** the backend system by successfully connecting the application with MongoDB for secure data management and Firebase for robust user authentication.
- **Integrated** machine learning models for real-time revenue forecasting and sentiment analysis, replacing static data with live backend API calls to ensure strict data isolation.

Remote Patient Monitoring System (RPMS)

- **Co-developed** a comprehensive healthcare management platform using Java (JDK 17), JavaFX, and MySQL Server 8.0, utilizing Maven for dependency management.
- **Designed** a multi-role architecture for Patients, Doctors, and Admins to manage user accounts, monitor vital signs, and handle emergency alerts.
- **Integrated** real-time communication tools, including chat and video consultations, alongside appointment scheduling, prescription management, and medical record uploads.

Product Listing Platform

- **Co-developed** a console-based product listing platform, utilizing file handling for data persistence across user accounts, product inventories, and transaction histories.
- **Designed** a dual-role authentication system with distinct login methods and privileges, enabling Admins to manage products and Customers to browse, leave reviews, and track order histories.
- **Engineered** core e-commerce functionalities, including custom sorting algorithms (Bubble and Selection Sort) for filtering, a persistent shopping cart, and simulated payment processing with automated receipt generation.

Technical Skills

Languages: Python, Java, Javascript, C++, SQL

Backend: FastAPI, Flask, REST API Design

Frontend: React (basic), Figma

Databases: MySQL, Firebase, MongoDB

Tools & DevOps: Git, GitHub, Docker (basic), Postman, VS Code

AI / ML: Machine Learning, Scikit-learn, MLOps

Education

2024 – present

Islamabad, Pakistan

Bachelor of Science in Artificial Intelligence

SEECs - NUST

Relevant Coursework: Machine Learning, Programming for AI, Data Structures & Algorithms, Database Systems, Computer Networks, Object-Oriented Programming

Courses

React Basics

Meta (Coursera)

Introduction to Cybersecurity

Cisco Networking Academy

Professional Experience

06/2025 – 08/2025

Project Intern

Murabbi 

- **Integrated** an advanced machine learning pipeline, combining a Whisper-based feature extractor with a BiLSTM + Attention model to deliver highly accurate phoneme predictions.
- **Engineered** secure and efficient backend communication by implementing Firebase Authentication and optimizing the system for high reliability and low-latency performance.
- **Designed** an interactive user interface featuring animated accuracy visualizations to provide users with immediate, engaging pronunciation feedback.

Extracurricular Activities

Sports & Leadership:

- **Crown FC:** Competed and represented the club across various All-Pakistan national-level football tournaments.
- **NUST Football:** Active collegiate player and team member

Community Service & Volunteering

- **NUST Community Serving Club (NCSC):** Active member contributing to university-led social welfare initiatives and ongoing community outreach programs.
- **Kawish NGO:** Dedicated volunteer for a student-driven non-governmental organization based in Islamabad, assisting with local community service projects.